

Direct laser sintering of metal powders: Mechanism, kinetics and microstructural features

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Received 31 December 2005; accepted 28 April 2006

Abstract

In the present work, the densification and microstructural evolution during direct laser sintering of metal powders were studied. Various ferrous powders including Fe, Fe–C, Fe–Cu, Fe–C–Cu–P, 316L stainless steel, and M2 high-speed steel were used. The empirical sintering rate data was related to the energy input of the laser beam according to the first order kinetics equation to establish a simple sintering model. The equation calculates the densification of metal powders during direct laser sintering process as a function of operating parameters including laser power, scan rate, layer thickness and scan line spacing. It was found that when melting/solidification approach is the mechanism of sintering, the densification of metals powders (D) can be expressed as an exponential function of laser specific energy input (ψ) as $\ln(1 - D) = -K\psi$. The coefficient K is designated as “densification coefficient”; a material dependent parameter that varies with chemical composition, powder particle size, and oxygen content of the powder material. The mechanism of particle bonding and microstructural features of the laser sintered powders are addressed.

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Keywords: Laser sintering; Rapid prototyping; Sintering rate; Metal powders; Microstructure

1. Introduction

Laser sintering is one of the leading commercial processes for rapid fabrication of functional prototypes and tools. The process creates solid three-dimensional objects by bonding powdered materials using laser energy. Different material systems such as engineering plastics, thermoplastic elastomers, metals, and ceramics are in use (<http://www.eos-gmbh.de>). Applications include patterns for investment casting, metal molds for injection molding and die casting, and molds and cores for sand casting [1–3]. Fabrication of prototype objects to enhance communication and testing of concepts during the design cycle are the other common usage of the process.

Although the laser sintering process can be applied to a broad range of powders, the scientific and technical aspects of the production route such as sintering rate and the effects of processing parameters on the microstructural evolution during the layer manufacturing process have not been well understood. This method of fabrication is accompanied by multiple modes of heat,

mass and momentum transfer, and chemical reactions that make the process very complex. As a consequent, the method essentially relies on empirical and experimental knowledge [4]. Typically, production of quality parts is dependent on the skill level and experience of the laser sintering machine operator. Therefore, future development of the process indispensably depends on the understanding of the densification mechanisms and the role of manufacturing parameters. The potential benefit from improved process understanding is related to development of intelligent process control and automation as well as to allow operators to select appropriate parameters value before processing [5].

The physical processes associated to laser sintering include heat transfer and sintering of powder. Recently, many works have been performed to develop computer models for the laser sintering process [6–9]. The models can be used as a tool to study how the various process variables affect the quality of the finished parts. The models usually take into account the thermal phenomenon involved in the process through differential equation for thermal diffusion. The thermal gradients in the powder bed are then related to thermal stresses, enabling predication of warpage or distortion of the fabricated parts. Nevertheless, to allow the evaluation of density as a function of time and tempera-

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Nomenclature

C	heat capacity (J/kg K)
d	beam diameter (mm)
D_n	particle size at n pct of the cumulative particle size distribution (μm)
h	scan line spacing (mm)
ΔH	latent heat of fusion (J/kg)
K	densification coefficient, Eq. (10)
k'	sintering rate
l	scan vector length (mm)
O	fraction of overlap, Eq. [3]
P	laser power (W)
$\bar{Q}$	laser energy density (J/mm ²)
$\bar{Q}$	delivered energy per volume of each laser track (J/mm ³)
S_w	particle size distribution slope
T	temperature (K)
v	scan rate (mm/s)
v^*	dimensionless velocity
w	layer thickness (mm)

Greek symbols

ε	void fraction
ρ	fractional density
τ	delay period between successive irradiation
ψ	laser energy input per volume of sintered specimen (kJ/mm ³)
η	coupling efficiency

Subscripts

b	bed
o	initial
s	sintering

ture and thus on the process variables, it is important to develop a sintering model. A sintering model allows one to perform a parametric analysis to study how variations in one parameter affect the sintered density or sintering depth within a powder bed.

In the indirect laser sintering process, i.e. selective laser sintering, which originated by the University of Texas at Austin, a polymer phase is used for powder particles bonding. During

laser irradiation, the polymer phase is melted and upon cooling, it bonds the powder particles together [10]. The sintering mechanism is viscous flow and the surface energy reduction drives the mass flow dissipation. Although modeling of laser sintering rates for real systems is difficult and in many cases may not provide accurate results, isothermal sintering rates of amorphous materials can be used. For instance, Nelson [11] assumed that the sintering rate is a function of temperature according to Arrhenius equation. Sun [12] related the sintering height to the laser power and laser velocity by simple empirical equations. Bugada et al. [13] considered the powder bed as an open pore network of cylinders and evaluated the sintered density as a function of time and temperature according to the Mackenzie–Shuttleworth model.

Unlike polymer materials, liquid phase sintering and melting/solidification approach are the mechanisms feasible for rapid bonding of powder particles in the direct laser sintering process (DMLS) [10]. So far, no much work has been performed to systematically investigate the sintering rate and densification of metal powders during DMLS. Since the process is very fast and complex, the existing isothermal sintering models are not adequate. For advanced theoretical and simulation studies, it is imperative to understand the sintering mechanism and kinetics. The effect of processing parameters on the microstructural features need to be established. This study is focused on parameters involved in determining the method of energy delivered to the powder medium, and to relate them to the sintering rate. Based on the empirical results, the mechanism of densification and microstructural evolution during DMLS are addressed. Although ferrous powders were used in the present work, the results are generic and can be applied for other metal powders.

2. Experimental

Table 1 gives the particle characteristics of iron and steel powders used as the starting materials. The particle size distribution was determined using a Coulter LS130 (Beckman Coulter GmbH, Krefeld, Germany) laser particle size analyzer. Except carbonyl iron powder, the other iron powders were produced by the water atomization process. In this case, sieving was used to obtain different particle sizes ranging from 10 to 200 μm . Graphite (2 μm , Aldrich), copper (<50 μm , Schlenk Metallpulver GmbH & Co. KG) and ferro-phosphorous (<45 μm) powders were blended with the iron powders in a Tumbling mixer for

Table 1
Particle size characteristics of laser sintered iron and steel powders

Powder	Supplier	Type (designation)	D_{10}	D_{50}	D_{90}	S_w
Iron	BASF Ludwigshafen, Germany	Carbonyl (CL)	4	10	29	3
	Quebec Metal Powders Limited, Quebec, Canada	Water atomized (Atomet 95)	11	26	54	3.7
	Quebec Metal Powders Limited, Quebec, Canada	Water atomized (Atomet 75)	20	41	68	4.9
	Hoeganaes AB, Hoeganaes, Sweden	Water atomized (ASC 100.29)	30	51	74	6.6
	Hoeganaes AB, Hoeganaes, Sweden	Water atomized (ASC 100.29)	34	68	108	5.1
	Hoeganaes AB, Hoeganaes, Sweden	Water atomized (ASC 100.29)	66	104	152	7.1
	Hoeganaes AB, Hoeganaes, Sweden	Water atomized (ASC 100.29)	128	171	227	10.2
	Hoeganaes AB, Hoeganaes, Sweden	Water atomized (ASC 100.29)	128	171	227	10.2
316L stainless steel	Ospery Metals Limited, UK	Gas atomized	7	19	38	3.6
M2 high-speed steel	Ospery Metals Limited, UK	Gas atomized	–	21	<45	–

30 min to obtain steels with different chemical compositions. Here, carbon steels (up to 1.2 wt% C) and ternary Fe–C–X (X = Cu, P) alloys were prepared. Prealloyed powders of 316L stainless steel and M2 high-speed steel (HSS) were also supplied from Osprey Metals Ltd., UK. The method of powder production was gas atomization. The carbonyl and gas atomized powders have near spherical shape whilst the water-atomized iron powders have irregular particles.

The prepared powders were sintered layer by layer to rectangular test specimens with dimensions of 10 mm × 10 mm × 7 mm using EOS M250X^{tend} machine (Electro Optical Systems GmbH, Germany). The detail of the laser sintering operation was described elsewhere [14]. The investigated laser sintering condition were laser power $P = 100\text{--}215\text{ W}$, scan rate $v = 50\text{--}600\text{ mm s}^{-1}$, layer thickness $w = 0.05\text{--}0.2\text{ mm}$ and scan

line spacing $h = 0.1\text{--}0.4\text{ mm}$. An alternative scanning pattern from layer to layer with equal line spacing in the X and Y directions was used. The diameter (d) of the laser beam was 0.4 mm. The building process was performed under nitrogen atmosphere and the parts were built on a low-carbon steel plate. The powder bed temperature was kept constant at 80 °C during laser sintering.

After removing the samples from the build plate, the density of the specimens was measured by the water displacement (Archimedes) and volumetric methods. Each processing condition was repeated at least twice and the result of the density measurement was expressed using the mean value. The standard deviation is less than 0.05 g/cm³. The surfaces of the as-sintered samples were observed in a LEO 438VP scanning electron microscopy (SEM). Samples for metallographic examination

Table 2
laser sintered density of the investigated powders

Material	Laser power (W)	Scan rate (mm s ⁻¹)	Layer thickness (mm)	Line spacing (mm)	Fractional density (%)
Fe	215	75	0.1	0.1	73.8
	192	75	0.1	0.1	73.8
	215	75	0.1	0.3	72.0
	192	75	0.1	0.3	71.0
	180	75	0.1	0.3	69.7
	162	75	0.1	0.3	68.5
	144	75	0.1	0.3	68.0
	125	75	0.1	0.3	67.4
Fe–0.8C	215	75	0.1	0.3	76.5
	192	75	0.1	0.3	75.0
	180	75	0.1	0.3	74.5
	162	75	0.1	0.3	73.1
	144	75	0.1	0.3	71.8
	125	75	0.1	0.3	70.0
	100	75	0.1	0.3	66.9
	215	50	0.1	0.3	78.1
	215	100	0.1	0.3	72.2
	215	125	0.1	0.3	71.4
	215	150	0.1	0.3	67.8
	215	200	0.1	0.3	64.2
	215	250	0.1	0.3	60.5
Fe–4Cu	215	75	0.1	0.3	74.9
	180	75	0.1	0.3	73.8
	144	75	0.1	0.3	70.7
	100	75	0.1	0.3	56.6
Fe–0.8C–4Cu–0.4P	215	75	0.1	0.3	80.6
	180	75	0.1	0.3	78.0
	144	75	0.1	0.3	75.0
	100	75	0.1	0.3	67.7
	166	300	0.1	0.3	59.0
	166	400	0.1	0.3	54.1
	166	500	0.1	0.3	51.3
	166	600	0.1	0.3	49.4
316L	215	50	0.05	0.3	93.6
	215	100	0.05	0.3	86.9
M2	200	50	0.1	0.3	88.2
	200	75	0.1	0.3	85.8
	200	100	0.1	0.3	84.5
	200	125	0.1	0.3	79.2
	200	150	0.1	0.3	76.6
	200	175	0.1	0.3	62.1

were prepared using standard techniques and etched in 2% Nital reagent (2 ml HNO_3 per 100 ml CH_3OH).

3. Results

In Table 2, the sintered density of investigated powders obtained at various laser sintering condition was summarized. The results reveal that the sintered density depends on both powder characteristics and the fabrication parameters. In general, as the energy input increases (higher laser power, P , or lower scan rate, v) higher density is obtained. Fig. 1 shows the fractional density of investigated powders as a function of P/v . To a first approximation, it seems that the density is linearly proportion to the ratio of laser power to scan rate in semi-log scale. From Table 2, one may also notice that the density depends on the layer thickness (w) and scan line spacing (h). Fig. 2 shows the effect of these parameters on the sintered density of iron powder with $D_{50} = 51 \mu\text{m}$. As it is seen, with increasing the layer thickness lower density is achieved. Also, with decreasing the line scan spacing higher density is obtained. Nevertheless, as it was shown elsewhere [15], at high laser energy inputs delamination of sintered layers and formation of large cracks are feasible (Fig. 3). This may lead to lower value of the sintered density measured. In conclusion, the results of the experiments determine a great influence of fabrication parameters on the densification of metal powders in DMLS process.

Other important factors which influence the densification during laser sintering are related to powder characteristics. As Table 2 shows, the amount of densification highly depends on the materials composition. Addition of elements such as carbon, phosphorous and copper enhances the sintering rate of iron powder at a given fabrication condition (Fig. 1). The results also indicate that the densification rate of high-alloy steels, e.g. 316L and M2, is more than that of iron and low-alloy powders processed at the same condition. This observation highlights the

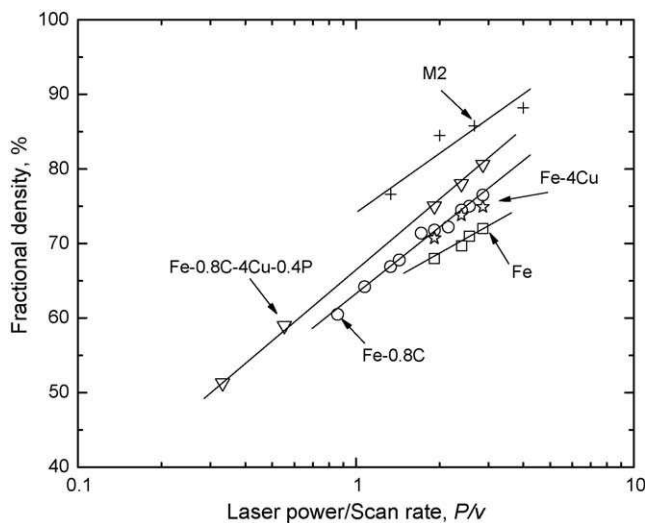


Fig. 1. Fractional density of laser sintered powders as a function of the ratio of laser power to scan rate. The layer thickness was 0.1 mm and the scan line spacing was 0.3 mm in all the experiments. The straight lines serve only as a guide to the eye.

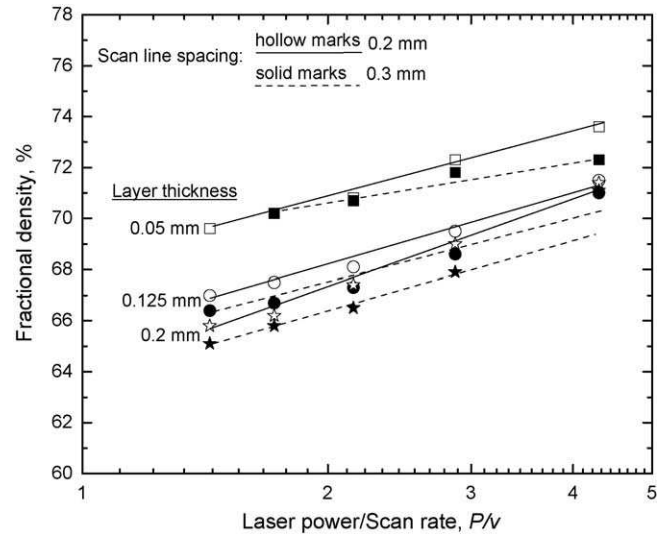
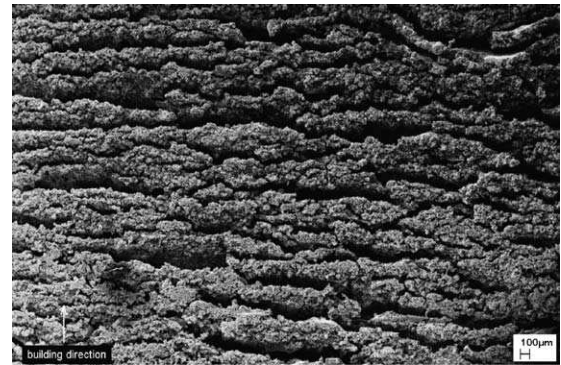
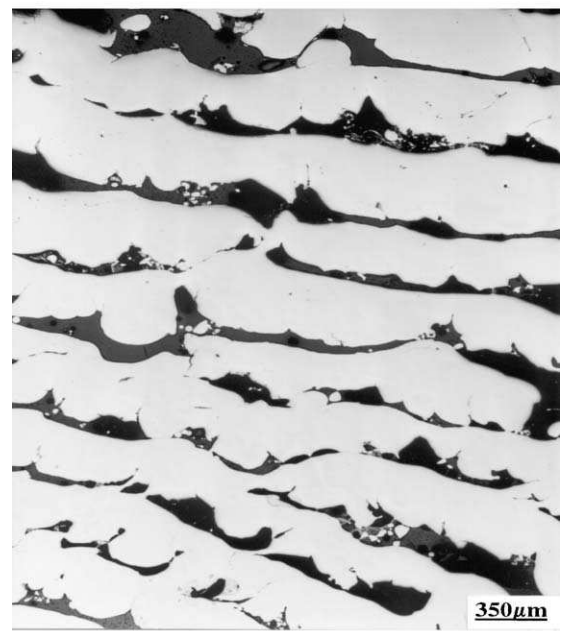


Fig. 2. Effect of layer thickness and scan line spacing on the fractional density of sintered iron ($D_{50} = 51 \mu\text{m}$) at laser power of 215 W. The straight lines only show the trend of variations.



(a)



(b)

Fig. 3. Delamination of the sintered layers (a) and formation of large horizontal cracks (b) in laser sintered iron ($D_{50} = 51 \mu\text{m}$) due to applying a high value of laser energy ($P = 215 \text{ W}$, $v = 50 \text{ mm s}^{-1}$, $h = 0.1 \text{ mm}$, $w = 0.05 \text{ mm}$).

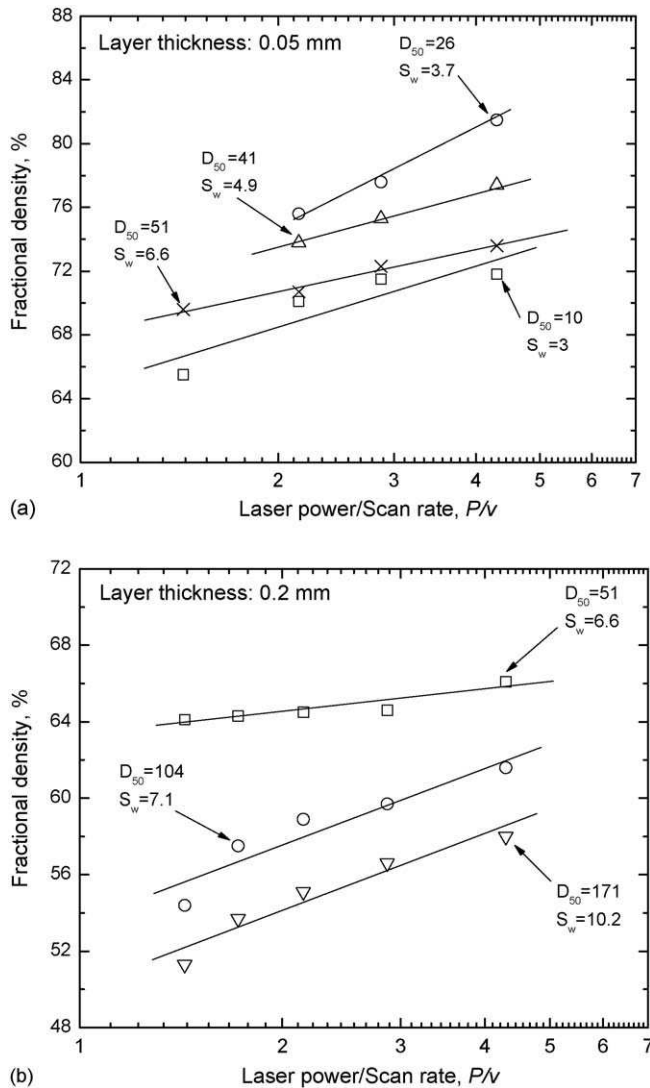


Fig. 4. Effect of powder particle size on the fractional density of laser-sintered iron at the scan line spacing of 0.2 mm. D_{50} is particle size at 50% of the cumulative particle size distribution (μm) and S_w is particle size distribution slope.

role of alloying elements on the particle bonding and thus on the sintering rate. On the other hand, the powder “size” (average size and particle distribution) is also of great importance. Fig. 4 shows the fractional density of various iron powders with particle sizes ranging from 10 to 200 μm . These specimens were laser sintered at different laser power and scan rate whilst the scan line spacing was 0.2 mm and the layer thickness was 0.05 mm (Fig. 4a) and 0.2 mm (Fig. 4b). Except iron powder with relatively fine particle size of 10 μm , it is apparent that the densification rate of finer powders is more than the coarser ones. In fact, the greater surface area of finer powders leads to higher sintering activity and thereby faster sintering rate. In this context, the effect of fabrication parameters should also be considered. When layer thickness of 0.05 mm was used, the effect of particle size seems to be more pronounced at higher laser energy inputs, i.e. higher values of P/v . In contrast, at layer thickness of 0.2 mm the impact of powder particle size on the densification appears more noticeably at lower laser energy inputs. The results determine that there is

a close relationship between the densification, the processing parameters and the powder characteristics.

4. Discussion

4.1. Beam-material interaction and mechanism of densification

It is known that when interaction of laser radiation with metal powders occurs, the energy deposition is performed by both bulk-coupling and powder-coupling mechanisms [4]. Note that multiple reflections effect between the powder particles leads to higher optical penetration depths compared to bulk materials. The density change during irradiation and formation of metal agglomerates also affect the coupling efficiency and thus influence the absorbed energy [15]. On the other hand, the effective thermal conductivity of the bed is lower than that of single particle and it highly depends on the amount of porosity, the arrangement of the particles and the contacts between them. In the practical range of scan rate in DMLS process, i.e. 50–2000 mm s^{-1} , exposure period of the laser irradiation (d/v) ranges between 0.2 and 8 ms whilst the radial thermal diffusion time ($d^2/4\alpha$) for a powder bed with thermal diffusivity of $(0.5\text{--}1) \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$ [4] is about 40–80 ms. In a such time scale, the heat flow distance during the interaction time is considerably less than the particle diameter, leading to very fast heating up the skin of the particles. The absorbed energy is then transferred to the surroundings by thermal diffusion. Therefore, the DMLS process can be considered as “high power density-short interaction time”. The temperature of the exposed powder particles can easily exceed the melting temperature, leading to full melting of the particles. Note that the energy intensity of the source might also be high enough to cause the material to evaporate [16]. Finite element analysis of the process presented elsewhere [17] for iron powder is shown in Fig. 5. This

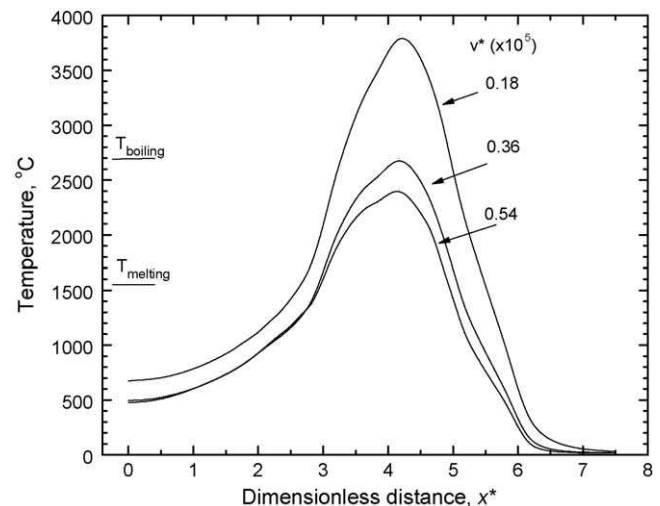


Fig. 5. Skin temperature distribution at the surface of iron powder for different dimensionless scan rate (v^*) calculated according to the three-dimensional heat flow equations for a moving Gaussian source with diameter of 0.4 mm on the surface of the powder bed. The coupling efficiency was assumed to be 20%.

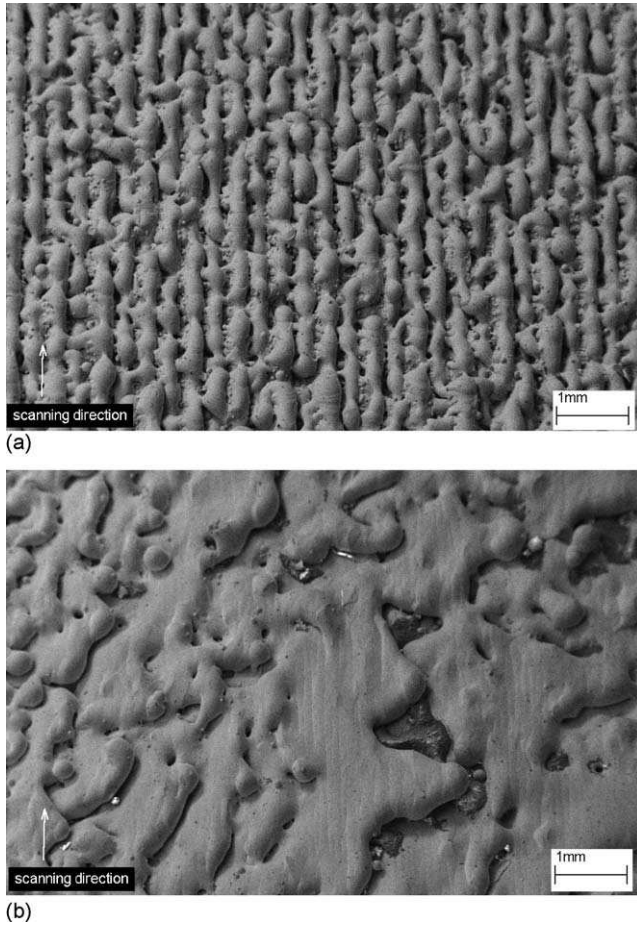


Fig. 6. SEM micrographs show the surface morphology of laser sintered iron. $P = 215 \text{ W}$, $v = 50 \text{ mm s}^{-1}$, $w = 0.1 \text{ mm}$, and $h = 0.3 \text{ mm}$ (a) and 0.1 mm (b).

graph illustrates the surface temperature distribution at different dimensionless scan rates defined according to the following equation:

$$v^* = \frac{vd}{4\alpha} \quad (1)$$

It can be noted that the temperature exceeds the melting point at high scan rates whereas at low scan rates, the temperature can be beyond the boiling point. Based on the experimental results reported previously [18], sound iron parts cannot be obtained at $v^* > 0.58$, determining that for particle bonding full melting must be occurred. The SEM micrographs taken from the surface of laser sintered specimen are shown in Fig. 6. The surface pictures clearly show that the powder particles were melted and formed a track of molten cylinder. The molten cylinder was then solidified to form continuous rows of metal agglomerates. The surface tension effect [19] and solidification shrinkage also resulted in the formation of large inter-agglomerates pores. Meanwhile, formation of metal balls due to instability of the molten cylinder according to Marangoni effect is likely to occur. The high thermal gradients present in the materials are accompanied by thermal stresses, which in fact may cause delamination of sintered layers (Fig. 3). Similar observations were noticed for Fe–C,

Fe–Cu, Fe–C–Cu–P, M2 HSS, i.e. full melting of the powder particles occurred (Fig. 7).

4.2. Densification rate

Many factors influence the densification of metal powders in DMLS process. In this paper, we classified these parameters in two categories as “process parameters” and “material properties”. Process parameters are the defined variables that influence and control the laser sintering process. These parameters are generally considered as those determine the amount of energy delivered to the surface. Material properties include chemical constitutions and the purity (oxygen, carbon, and nitrogen) of the material, method of alloying (elemental or prealloy) and particle characteristics (size, distribution and shape). Assuming fixed material properties, the amount of delivered energy during the laser irradiation-material interaction period is dependent on the intensity of the laser beam, the period of a single exposure, the number of total exposures and the time between each exposure. The parameters that are most influential in governing the intensity and method of the energy delivered to a single layer of powder with thickness (w) are the laser power (P), laser beam spot size (d), scan rate (v), scan line spacing (h) and geometrical scanning strategy including the length of scan vector and the method of irradiation between each successive layers. The effect of scanning method will be addressed in the following section. Removing any geometrical constraint, the energy density can be expressed as:

$$Q = \frac{\pi\eta P}{4dv} \quad (2)$$

Since DMLS process exposes a region to the laser beam irradiation several times by following an overlapping scanning pattern, the fraction of overlap is simply defined as:

$$O = \frac{d}{h} \quad (3)$$

Therefore, the delivered energy per volume of each laser track is given by:

$$\bar{Q} = \frac{\pi\eta P}{4hvw} \quad (4)$$

This energy rapidly heats up the particles above the melting point and cause particle bonding to occur. Therefore, the apparent sintering temperature (T_s) in very simple form is related to the relevant process parameters as:

$$T_s = T_o + \frac{1}{C} \left[\left(\frac{\pi\eta}{4\rho} \right) \left(\frac{P}{hvw} \right) - \Delta H \right] \quad (5)$$

Note that if material evaporation occurs, the latent heat of evaporation must also be considered in this equation. In this article, we define the specific energy input (ψ) as the total energy input per volume of each sintered track according to Eq. (6).

$$\psi = \frac{P}{hvw} \quad (6)$$

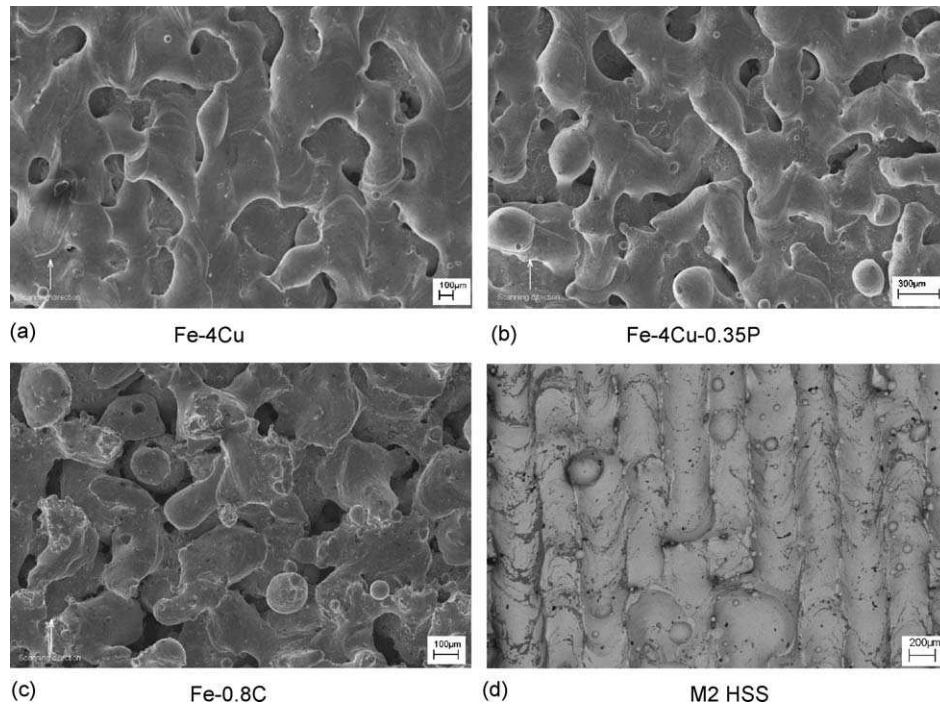


Fig. 7. SEM micrographs show the surface morphology of laser sintered steels. These specimens were produced at $P=215$ W, $v=75$ mm s⁻¹, $h=0.3$ mm and $w=0.1$ mm.

Therefore, it can be deduced that ψ is in direct relation to T_s and thus to the sintering rate. One may use the available general sintering rate equations [20] which have been suggested for liquid phase sintering technology in order to relate the densification rate to the working temperature. In the present work, the rate changes in void fraction of powder bed in DMLS process are simply assumed to flow the *first order* kinetics law according to the following equation:

$$\frac{\partial \varepsilon}{\partial t} = -k' \varepsilon \quad (7)$$

Fig. 8 shows the porosity of laser sintered iron as a function of exposure period of the laser irradiation (t). As it is seen, the experimental results are in good agreement with the sintering kinetics model. However, it should be noted that the sintering rate (k') depends on P/wh . According to the results presented in Fig. 9, it is apparent that the sintering rate, in a first approximation, is linearly proportional to P/wh . It is also important to note that in Eq. (7), the boundary condition is as follows:

$$\varepsilon_b \leq \varepsilon \leq \varepsilon_s \quad (8)$$

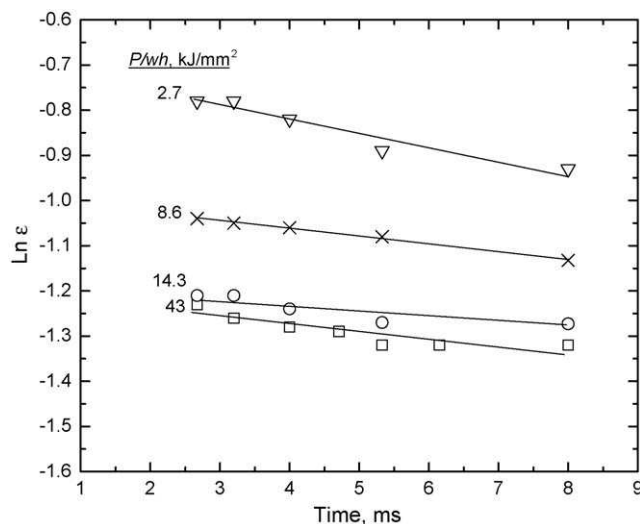


Fig. 8. Void fraction of laser sintered iron ($D_{50}=51$ µm) versus exposure time of laser irradiation (CO₂ laser beam with 0.4 mm diameter).

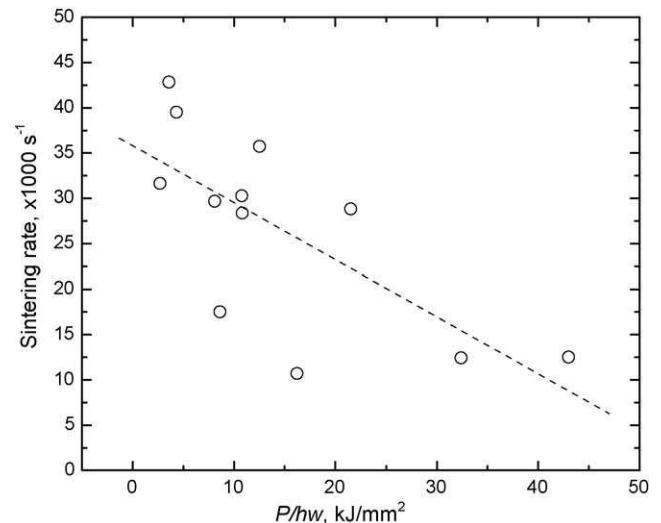


Fig. 9. Sintering rate data of iron powder ($D_{50}=51$ µm) in DMLS process. The straight line serves only as a guide to the eye.

where ε_b is the initial void fraction of the powder bed before the start of laser sintering and ε_s is the minimum attainable porosity in a sintered part. It is noteworthy that even at very intensive laser energy full densification cannot be obtained because of delamination of the sintered layers due to thermal stresses (Fig. 3), formation of gas pores during solidification [21] and porosity formation due to material shrinkage and the balling effect (Figs. 6 and 7). The value of ε_s depends on the materials properties and changes between 0.02 and 0.3. By defining the densification factor as

$$D = \frac{\varepsilon - \varepsilon_b}{\varepsilon_s - \varepsilon_b} \quad (9)$$

the amount of densification can be related to the specific energy input such that

$$\ln(1 - D) = -K\psi \quad (10)$$

where K is designated as “densification coefficient”. It is noteworthy that this relation is only valid for a condition, in which particle bonding is carried out by full melting/solidification approach, i.e. powder melting.

To verify the sintering equation, some experiments were performed. Fig. 10 shows plots of “ $-\ln(1 - D)$ versus ψ ” for the examined materials according to the experimental data. The straight lines yield the correlation constant K , which in turn, higher value means less densification during the laser sintering process. As it can be seen, this constant strongly depends on the materials examined. For instance, addition of 4 wt.% copper to iron powder reduced the K value from 18.1 to 11.9 (Fig. 10a), meaning that the densification was improved, in agreement with the experimental results reported previously [14]. Similar trend can be noticed for Fe–0.8%C system in compatible with the role of graphite on the laser sintering of iron powder as presented elsewhere [22]. From Fig. 10b, it is also apparent that M2 High-speed steel powder is not densified significantly during laser sintering, i.e. the K value is relatively high. Akhtar [23] and Wright [24] have shown that this powder is not suitable for processing because of low attainable density (67% theoretical). Many other reports can be found in literature, for example in [25,26], concerning DMLS of 316L stainless steel, demonstrating that the processability of this powder is very good in accordance with the low K value obtained.

It is known that finer particles provide a larger surface area to absorb more laser energy, which increases the working temperature and thus the sintering kinetics [27]. Therefore, it is expected that laser sintering of finer particles tends to enhance the kinetic of densification. Fig. 11 shows the effect of particle size on the densification coefficient of iron powder. It is seen that coarser powders show lower densification (higher K value) in the course of laser sintering. This relationship holds true for the other materials examined. For instance, the densification plot of Fe and Fe–0.8C–4Cu–0.35P powders with two particle sizes of <50 and <100 μm is shown in Fig. 12. It is noticeable that the influence of alloying elements on the densification highly depends on the particle size used. In the case of coarse powder (<100 μm) the densification is marginally influenced by the addition of carbon, copper and phosphorus. In contrast, when

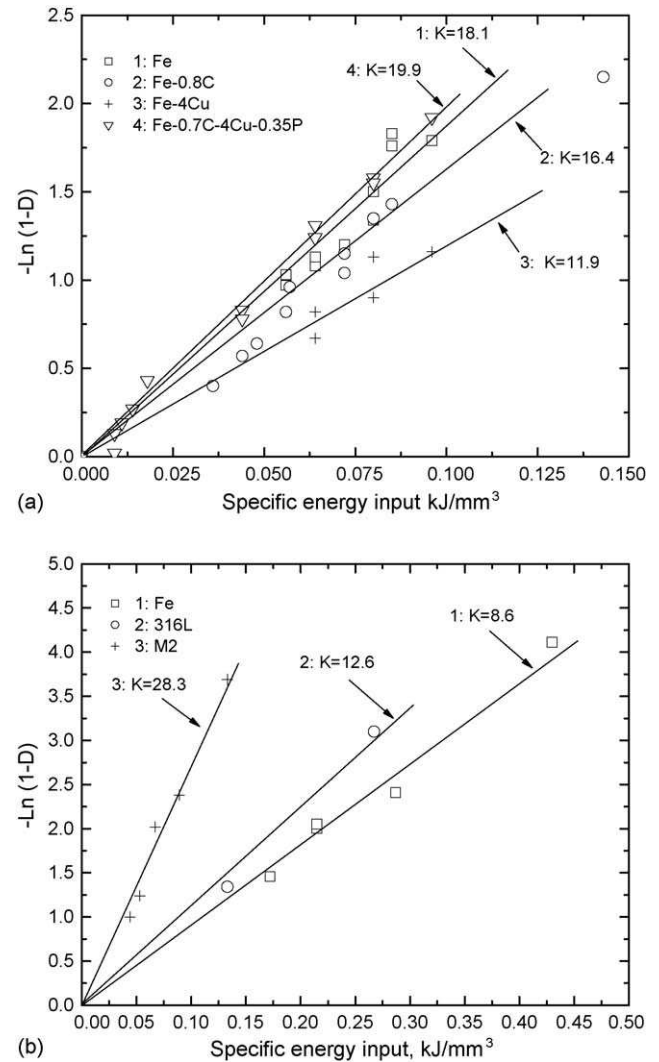


Fig. 10. The densification (D) of metal powders as a function of the specific energy input (ψ) for a continuous CO_2 laser source with diameter of 0.4 mm. The particle size of the powders were <100 μm (a) and <45 μm (b).

the finer powder is used, the K value decreased from 12.1 to 4.4, showing a significant improvement in densification by the alloying elements.

4.3. The effect of processing parameters

The aforementioned sintering equation simply relates the densification of metal powders to the most influencing parameters of the laser sintering process. These parameters include the laser energy power, scan rate, layer thickness, and scan line spacing. The effect of material properties such as chemical constitutions, particle size and particle distribution can be considered in the defined densification coefficient, K . Nevertheless, there are some other factors such as scanning strategy and sintering atmosphere that influence the sintering rate. To understand the importance of these parameters few experimental runs were performed. Fig. 13 illustrates the effect of scanning vector length (l) and sintering atmosphere on the fractional density of laser sintered iron. For evaluation of the influence of scan vec-

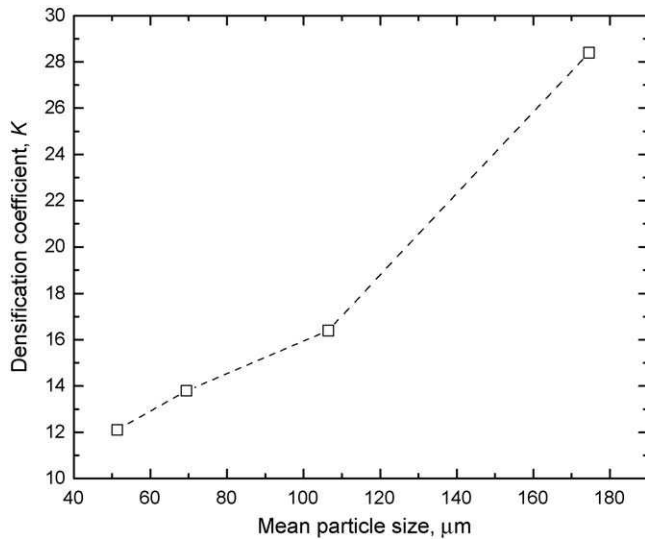


Fig. 11. Effect of powder particle size on the densification coefficient of water atomized iron powder with oxygen content of 0.07 wt%. Higher K value shows lower densification during direct laser sintering. The dash line only serves as a guide to the eye.

tor length, rectangular specimens with varying length to wide ratio were produced. The results of density measurement are presented in Fig. 13a. It is noticeable that the sintered density slightly decreases as the vector length increases. For instance, when the scan length was increased from 10 to 60 mm, about 6% lower fractional density has been obtained. This observation can be explained according to the delay period (τ) between successive irradiation defined as

$$\tau = \frac{l}{v} \quad (11)$$

As the vector length increases, the higher delay period between successive irradiation leads to a decrease in the amount of energy stored on the surface. Consequently, less densification is expected during the laser sintering. Another consequence of

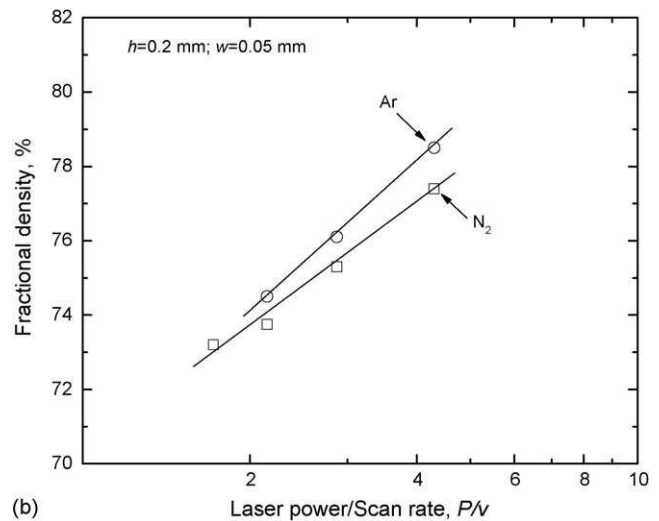
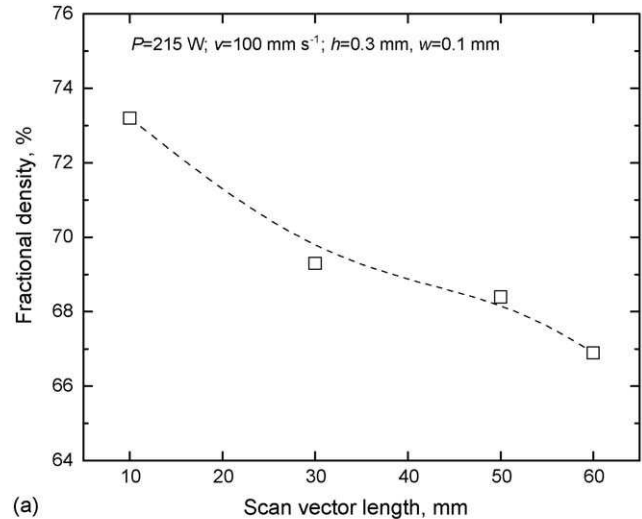


Fig. 13. Effect of scan vector length (a) and sintering atmosphere (b) on the fractional density of laser sintered iron ($D_{50} = 41 \mu\text{m}$).

applying higher vector length is related to the development of thermal stresses. It is known that an increase in the scan length results in development of higher thermal stresses in the sintered parts [28,29]. These stresses are responsible for part cracking, warpage and Christmas tree defects.

The influence of sintering atmosphere on the densification of iron powder is shown in Fig. 13b. It appears that sintering in an argon atmosphere yields slightly better densification rather than nitrogen. This observation is in accordance with the findings presented previously [30,31] for M2 high-speed steel, i.e. sintering under nitrogen atmosphere yields less densification at low scan rates whilst at high scan rates, the influence of sintering atmosphere is marginal. Nevertheless, the effect of sintering atmosphere as compared to the other parameters is not very pronounced. The dissolution of nitrogen in the iron melt during the laser processing may serve some influence. The impact of sintering atmosphere on the densification could also be related to the amount of oxygen present during sintering. It is known that the presence of oxygen allows surface oxides and slags to form as the powder particles are heated and melted by the scanning

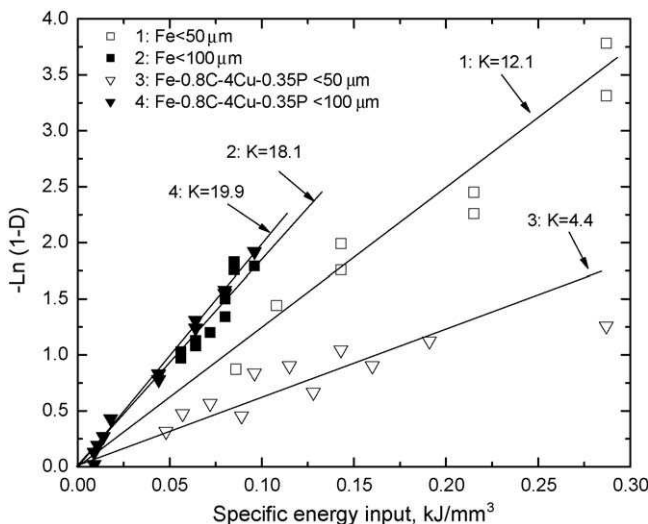


Fig. 12. Effect of particle size on the densification of Fe and Fe-0.8C-4Cu-0.35P powders.

laser beam [32]. The formation of oxide layer on the surface of powder particles significantly increases the absorption rate of CO₂ laser radiation [33]. This changes the temperature-time history of sintering and increases the melt volume, allowing surface tension become more dominant. Another concern is the liquid metal surface tension, which influences the wetting angle between the solid and the liquid phases that can disrupt bonding between rastered lines and individual layers [10,34]. Therefore, the amount of oxygen present during the heating, melting, and fusion of metal powders in the laser sintering process should influence the densification and the attendant microstructural features. As it has been addressed in the previous publication [30], the effect of oxygen can be considered in the densification coefficient, K . Anyway, except the role of oxygen that can be taken into account by using the densification coefficient, the effect of protecting gas is not very pronounced and does not impact the sintering rate considerably.

5. Conclusions

The effect of processing parameters on the densification of metal powders in the laser sintering process was investigated. Based on the empirical sintering rate data, a relationship was established between the densification of metal powders during DMLS and the energy delivered to the powder bed by the laser beam. This relationship has been shown to be useful for metals with congruent melting point or alloys, which full melting/solidification approach is feasible mechanism of rapid particle bonding in DMLS process. The following conclusions can be afforded.

1. The laser sintering can be considered as “high power density-short interaction time” process. The delivered energy heats up the exposed powder particles rapidly beyond the melting temperature. The particle bonding is then performed and the kinetics of densification depends on the working temperature. Consequently, the parameters involved in determining the method of energy delivered to the powder medium control the sintering rate. Meantime, the evaporation of exposed powder in the laser sintering process may occur, particularly at an intensive laser energy input.
2. It was found that as the laser energy input increases (higher laser power; lower scan rate; lower scan line spacing; lower layer thickness) better densification is achieved. Nevertheless, there is a saturation level, in which, even at very intensive laser energy full density cannot be obtained.
3. When melting/solidification approach is the mechanism of densification, the rate changes in void fraction of powder bed in DMLS process obeys the *first order* kinetic law: $\partial\epsilon/\partial t = -k'\epsilon$. The sintering rate (k') was found to be a function of the laser energy input. Therefore, the sintered density of metal powders in DMLS process should be an exponential function of the laser energy input.
4. Besides the fabrication parameters, the powder properties strongly influence the densification kinetics. Finer particles provide larger surface area to absorb more laser energy, leading to a higher sintering rate. The chemistry and the shape

of the particles also affect the densification in DMLS process. To take into account the material characteristics in the sintering model, a densification coefficient (K) was defined and used. It was shown that this coefficient is related to the powder characteristics (chemical composition, particle size, particle size distribution, oxygen content, etc.) and the sintering rate increases as the K value decreases.

5. The results showed that laser scanning strategy and sintering atmosphere influence the densification. Although the development of thermal stresses highly depends on the scanning strategy, the effect of scan vector length on the sintered density was found to be marginal. If the oxygen potential of the sintering atmosphere is kept constant, the role of protective atmosphere on the densification is also not very pronounced.

Acknowledgements

The laser-sintered specimens were prepared at Fraunhofer Institute for Manufacturing and Advanced Materials (IFAM), Bremen, Germany. The help of Dr. F. Petzoldt and Dr. H. Pohl is gratefully acknowledged. The grant of the Office of Vice President for Research and Technology, Sharif University of Technology, is sincerely appreciated.

References

- [1] J. Haenninen, DMLS Moves from Rapid Tooling to Rapid Manufacturing, Metal Powder Report, September 2001, pp. 24–29.
- [2] R. Jiang, W. Wang, J.G. Conley, in: Proceedings of the Solid Freeform Fabrication Symposium on Application of FFF in the Metal Casting Industry, Compiled by D.L. Bourell, J.J. Beaman, R.H. Crawford, H.L. Marcus, J.W. Barlow, The University of Texas at Austin, TX, 1999, p. 767.
- [3] T. Wohlers, Rapid Prototyping and Tooling Worldwide: Stalled Growth, Countless Benefits, Vast Confusion, CATIA Solution Magazine, January/February 2000.
- [4] P. Fischer, V. Romano, H.P. Weber, N.P. Karapatis, E. Boillat, R. Glardon, Acta Mater. 51 (2003) 1651–1662.
- [5] J.D. Williams, C.R. Deckard, Rapid Prototyping J. 4 (2) (1998) 90–100.
- [6] Y. Zhang, A. Faghri, Int. J. Heat Mass Transfer 42 (1999) 775–788.
- [7] S. Kolossov, E. Boillat, R. Glardon, P. Fischer, M. Locher, Int. J. Machine Tools Manufact. 44 (2004) 117–123.
- [8] K. Dai, L. Shaw, Acta Mater. 52 (2004) 69–80.
- [9] H. Chung, S. Das, Int. J. Heat Mass Transfer 47 (2004) 4153–4164.
- [10] D.L. Bourell, H.L. Marcus, J.W. Barlow, J.J. Beaman, Int. J. Powder Metall. 28 (4) (1992) 369–381.
- [11] J.C. Nelson, Selective laser sintering: a definition of the process and an empirical sintering model, Ph.D. dissertation, University of Texas, Austin, TX, 1993.
- [12] M.M. Sun, Physical modeling of the selective laser sintering, Ph.D. dissertation, University of Texas, Austin, TX, 1991.
- [13] G. Bugeda, M. Cervera, G. Lombera, Rapid Prototyping J. 5 (1) (1999) 21–26.
- [14] A. Simchi, F. Petzoldt, H. Pohl, J. Mater. Process. Technol. 141 (3) (2003) 319–328.
- [15] Y. Kizaki, H. Azuma, S. Yamazaki, H. Sugimoto, S. Takagi, Jpn. J. Appl. Phys. 32 (1A) (1993) 205–212.
- [16] J.P. Kruth, L. Froyen, J. Van Vaerenbergh, P. Mercelis, M. Rombouts, B. Lauwers, J. Mater. Process. Technol. 149 (2004) 319–328.
- [17] A. Simchi, Direct Laser Sintering of Iron Base Powders, Presented in Fraunhofer Institute IFAM, Bremen, Germany, March 2001.
- [18] A. Simchi, H. Pohl, Mater. Sci. Eng. 359A (1–2) (2003) 119–128.

- [19] S. Das, in: *Proceedings of the Solid Freeform Fabrication Symposium on some Physical Aspects of Control in Direct Selective Laser Sintering of Metals—Part III*, Compiled by D.L. Bourell, J.J. Beaman, R.H. Crawford, H.L. Marcus, J.W. Barlow, The University of Texas at Austin, 2001, pp. 102–109.
- [20] R.M. German, *Liquid Phase Sintering*, Plenum Press, NY, 1985.
- [21] M. Rombouts, L. Froyen, J.-P. Kruth, J.V. Vaerenberg, P. Mercelis, in: *Proceedings of the Powder Metallurgy World Congress on Production and Properties of Dense Iron Based Parts Produced by Laser Melting with Plasma Formation*, Euro PM2004, Compiled by H. Danninger, R. Ratzl, vol. 5, Vienna, 2004, pp. 115–121.
- [22] A. Simchi, H. Pohl, *Mater. Sci. Eng. A* 383 (2004) 191–199.
- [23] S. Akhtar, C.S. Wright, M. Youseffi, C. Hauser, T.H.C. Childs, C.M. Taylor, in: *Proceedings of the European Powder Metallurgy Conference on Direct Selective Laser Sintering of Tool Steel Powders to High Density*, Euro PM2003, EPMA, Schremsbury, vol. 3, Valencia, 2003, pp. 379–384.
- [24] C.S. Wright, S.P. Akhtar, M. Youseffi, C. Hauser, T.H.C. Childs, J. Xie, P. Fox, in: *Proceedings of the Powder Metallurgy World Congress on Laser Re-melting of Prealloyed Steel Powders to High Density*, Euro PM2004, Compiled by H. Danninger, R. Ratzl, vol. 5, Vienna, 2004, pp. 109–114.
- [25] A.J. Pinkerton, L. Li, *Thin Solid Film* 453–454 (2004) 600–605.
- [26] W. O'Neill, C.J. Sutcliffe, R. Morgan, K.K. B. Hon, in: *Proceedings of the Solid Freeform Fabrication Symposium on Investigation of Short Pulse Nd:YAG Laser Interaction with Stainless Steel Powder Beds*, Compiled by D.L. Bourell, J.J. Beaman, R.H. Crawford, H.L. Marcus, J.W. Barlow, The University of Texas at Austin, 1998, pp. 147–159.
- [27] H.J. Niu, I.T.H. Chang, *Scripta Mater.* 41 (1) (1999) 25–30.
- [28] X.C. Li, J. Stampfl, F.B. Prinz, *Mater. Sci. Eng. A* 282 (2000) 86–90.
- [29] A.H. Nickel, D.M. Barnett, G. Link, F.B. Prinz, in: *Proceedings of the Solid Freeform Fabrication Symposium on Residual Stresses in Layered Manufacturing*, Compiled by D.L. Bourell, J.J. Beaman, R.H. Crawford, H.L. Marcus, J.W. Barlow, The University of Texas at Austin, 1999.
- [30] C. Hauser, T.H.C. Childs, K.W. Dalgarno, R.B. Eane, in: *Proceedings of the Solid Freeform Fabrication Symposium on Atmospheric Control during Direct Selective Laser Sintering of Stainless Steel 314S Powder*, Compiled by D.L. Bourell, J.J. Beaman, R.H. Crawford, H.L. Marcus, J.W. Barlow, The University of Texas at Austin, 1999, pp. 265–272.
- [31] N. Tolochko, S.E. Mozzharov, N.V. Sobolenko, Yu.V. Khlopkov, I.A. Yadroitsev, V.B. Mikhailov, *J. Adv. Mater.* 2 (2) (1995) 151–157.
- [32] A. Simchi, *Metal. Mater. Transact.* 35B (2004) 937–948.
- [33] A. Simchi, H. Asgharzadeh, *Mater. Sci. Technol.* 20 (11) (2004) 1462.
- [34] H. Asgharzadeh, A. Simchi, *Mater. Sci. Eng. A* 403 (1–2) (2006) 290–298.